Wireless connection

The marriage of wireless and networks has seen the evolution of WLANs that have enjoyed a remarkable success in the LAN world during the last decade.

WLAN technology has been adopted at two levels. The first, and the most commonly deployed, form co-exists with the wired network at the workplace to give users mobility and ease-of-reconfiguration.

The second level is its large-scale deployment through the setting up of hotspots—geographic locations where an access point provides wireless broadband network services to mobile visitors.

While vendors are doing their bit to push wireless among potential users, there are still issues such as security, spectrum availability and above all, lack of awareness retarding its growth.

These hurdles notwithstanding, wireless is undeniably a technology that holds tremendous potential and resellers would do well to keep themselves abreast of its basics and the advancements in this area.

Another point in its favour, as far as India is concerned, is that there's no license fee on wireless LAN implementation. The headache of getting permission is taken up by the device manufacturer.

Plug these PCMCIA cards into your laptop to go wireless

As far as IEEE networking standards are concerned, the Indian Government's Wireless Planning Commission (WPC) has permitted unlicensed usage of the IEEE 802.11b wireless networking protocol that allows for Wi-Fi connectivity at 11 Mbps. The same is not the case with the IEEE 802.11g standard: The regulations are not very clear but vendors claim that implementation within a closed environment are possible. You'd do well to check on the current status on the regulation.

Wi-Fi hacker alert!

Security is a critical issue with wireless transmission. Since the signal in a WLAN is freely transmitted in space, anybody in the vicinity has access to it. This can include a hacker sitting in a car parked just outside your home or office. SSIDs are used to identify the wireless adapters installed in a wireless environment, by access points. The SSID is broadcasted freely many times by an access point. A hacker with reasonable experience and the help of an 802.11 analysis tool, can sniff the network to get the SSID, and hence, free access to the home or office network.

Wireless networks can enable WEP (Wired Equivalent Privacy), a form of encryption, available in 64, 128 and 256- bit that will keep amateur hackers away. However, an experienced hacker can gain access by monitoring the data transmission and decoding the WEP-encrypted information. WPA (Wireless Protected Access), a more secure encryption standard, is in under development, and will be available soon.